

Research Agendas Gender, AI, and the Economics of Peer Review

Three Integrated Research Agendas

Using the ResearchAI Development Suite

AI-Powered Multi-Agent Peer Review System

May 2026

Draft — Not for Circulation

Contents

Executive Summary	5
Research Agenda 1	7
1 Motivation and Research Question	7
2 Contribution to the Literature	8
3 Data	9
3.1 Paper Corpus	9
3.2 Author Classification	10
3.3 AI Quality Evaluation (ResearchAI)	10
3.4 NLP-Extracted Rhetorical Features	10
3.5 Readability Measures	11
4 Empirical Strategy	11
4.1 Step 1: Baseline Balance Test (Working Papers)	11
4.2 Step 2: The Delta Score — Quality Change During Peer Review	12
4.3 Step 3: Published Quality Conditional on Baseline	13
4.4 Step 4: Rhetorical Mediators	13
4.5 Step 5: The Intersectional Penalty (Gender $\times$ Global South)	14
4.6 Step 6: Decomposition	15
4.7 Step 7: Career Trajectory Analysis	16
5 Power Analysis and Sample Size Determination	17
5.1 Effect Size Calibration	17
5.2 Sample Size Calculations	17
5.3 Multiple Testing Adjustment	18
5.4 Pilot Protocol	18
6 Identification Challenges and Robustness	18
7 Validation: AI vs. Human Evaluation	19
8 Expected Results and Policy Implications	19
9 Literature Review Strategy	19
9.1 The Five Literature Strands	19
9.1.1 Strand 1: Gender Bias in Peer Review	19
9.1.2 Strand 2: AI and Algorithmic Evaluation in Expert Settings	20

9.1.3	Strand 3: Gendered Academic Discourse and Rhetoric	20
9.1.4	Strand 4: Intersectionality in Economics	20
9.1.5	Strand 5: Discrimination and Information	20
9.2	Critical Literature Gap: The 2×2 Matrix	20
10	Timeline	21
Research Agenda 2		22
11	Motivation and Research Question	22
12	Contribution to the Literature	23
13	Experimental Design	23
13.1	The Three-Layer Audit Framework	23
13.2	Paper Sample	24
14	Empirical Strategy	24
14.1	The Audit Test: Does Author Identity Affect AI Scores?	24
14.2	Does AI Bias Mirror Human Bias?	25
14.3	Implicit Gender Detection Test	26
14.4	Text Analysis of AI Reviews	26
14.5	Debiasing Experiments	27
15	Connection to the Research Program	27
15.1	The AI as Benchmark for Agenda 1	27
15.2	Calibrating Human vs. AI Bias	28
16	Literature Review Strategy	28
16.1	The Five Literature Strands	28
16.1.1	Strand 1: AI Fairness and Algorithmic Bias (Primary)	28
16.1.2	Strand 2: Correspondence Study Methodology	29
16.1.3	Strand 3: Gender Bias in Peer Review	29
16.1.4	Strand 4: AI Adoption in Academia (Urgency)	29
16.1.5	Strand 5: Debiasing Interventions for AI	29
16.2	Critical Week-1 Confirmation Search	29
17	Timeline	29

Research Agenda 3	30
18 Motivation and Research Question	30
19 Contribution to the Literature	31
20 The Natural Experiment: Timeline and Identification	32
20.1 Key Dates	32
20.2 Identification Strategy	32
21 Data	32
21.1 Working Paper Corpus with Time Variation	33
21.2 AI Adoption Intensity Proxy	33
22 Empirical Strategy	33
22.1 Step 1: Difference-in-Differences on Working Paper Quality	33
22.2 Step 2: Event Study	34
22.3 Step 3: Diff-in-Diff on the Quality Delta	35
22.4 Step 4: Adoption Intensity and Heterogeneity	36
22.5 Step 5: Global South Heterogeneity	36
22.6 Step 6: Rhetorical Convergence	37
23 Identification Challenges and Robustness	37
23.1 Threats to Identification	37
23.2 Robustness Checks	38
24 Power Analysis	38
24.1 Effect Size Calibration	39
24.2 Sample Size Adequacy	39
25 Expected Results and Scenarios	39
26 Literature Review Strategy	40
26.1 The Five Literature Strands	40
26.1.1 Strand 1: AI as Equalizer in the Workplace	40
26.1.2 Strand 2: Gender Gaps in Peer Review	40
26.1.3 Strand 3: Technology Adoption and the Digital Divide	40
26.1.4 Strand 4: Natural Experiments from Technology Shocks	40
26.1.5 Strand 5: AI Detection and Text Analysis	40
26.2 Critical Differentiation from Existing Work	41
27 Feasibility and Cost	41

28 Timeline	42
How the Three Agendas Connect	43
References	45

Executive Summary

This document presents three integrated research agendas that extend the foundational work of [Hengel \(2022\)](#) and [Alexander et al. \(2023\)](#) on gender bias in academic peer review. Each agenda leverages the ResearchAI Development Suite—an AI-powered, multi-agent peer review system—and incorporates a **Global South dimension** that is absent from the existing literature.

Agenda 1: The AI Counterfactual. Uses gender-blind AI evaluation as a multi-dimensional quality benchmark. Evaluates working papers at baseline, tests gender balance in AI scores, and measures the delta (published minus working paper) across 12 quality dimensions to reveal where peer review demands more from women. Incorporates rhetorical and argumentative analysis as mediating channels, and tests whether the gender tax compounds at the intersection of gender and Global South origin.

Agenda 2: Does the AI Referee Discriminate? An audit study sending identical papers with swapped author identities—varying gender, ethnicity, and institutional affiliation—through multiple LLMs. Tests whether AI replicates, amplifies, or mitigates human biases, including geographic discrimination. Validates the AI tool used in Agenda 1.

Agenda 3: Did AI Tools Close the Gender Gap? Exploits the release of ChatGPT (November 2022) and GPT-4 (March 2023) as a natural experiment, comparing the gender gap in paper quality before and after AI writing tools became available. Tests whether AI served as an equalizer or an inequality amplifier, with particular attention to whether benefits reached Global South researchers. Decomposes the gender gap into supply-side (manuscript quality) and demand-side (reviewer treatment) components without requiring an RCT.

The three agendas form a coherent research program: Agenda 2 validates the AI tool, Agenda 1 deploys it to document the gender tax, and Agenda 3 tests whether the generative AI revolution has begun closing the gap. The Global South dimension runs through all three, ensuring that findings generalize beyond the Western institutions studied by [Hengel \(2022\)](#) and [Alexander et al. \(2023\)](#). Critically, all three agendas share the same data infrastructure and can proceed without participant recruitment, IRB approval, or external funding.

Table 1: How Prior Proposals Map to the Three Agendas

Agenda	Incorporates
Agenda 1: AI Counterfactual	Former Proposal 1 (core AI counterfactual) + Proposal 2 (rhetorical analysis as mediators) + Proposal 4 (intersectional gender $\times$ geography heterogeneity)
Agenda 2: AI Bias Audit	Former Proposal 5, enhanced with geographic/ethnic name variation
Agenda 3: AI Revolution	Former Proposal 3 (RCT $\rightarrow$ natural experiment using ChatGPT/GPT-4 release as exogenous shock)

Research Agenda 1

The AI Counterfactual — Measuring the Gender Tax in Peer Review Using Gender-Blind Algorithmic Evaluation

Target Journals: *Quarterly Journal of Economics*, *Review of Economic Studies*, *American Economic Review*

1 Motivation and Research Question

Hengel (2022) established that female-authored papers in top economics journals are 1–6% more readable than male-authored papers, and Alexander et al. (2023) documented that female-authored papers spend 29–60 additional days under review at *Energy Economics* and 18–29 days at 32 other Elsevier journals. Both papers rely on *indirect* measures of quality—readability formulas and citation counts—to argue that the gender gap in peer review outcomes is not driven by quality differences.

This paper proposes a fundamentally different approach. We use a multi-agent AI peer review system (ResearchAI) that evaluates papers *holistically*—across empirical strategy, theoretical contribution, writing quality, data transparency, and methodological rigor—without any information about author identity. Rather than requiring hard-to-obtain review outcome data (review duration, number of revision rounds), we exploit the existence of **matched working paper–published paper pairs** to measure how much peer review changes paper quality across 12 dimensions, and whether these changes differ by gender.

Our approach has three layers. First, we test whether male- and female-authored working papers start at the same AI-assessed quality level (baseline balance). Second, we measure the *delta*—the quality change from working paper to published version—across all 12 dimensions, testing whether women’s papers must improve more to get published. Third, we test whether these patterns are amplified at the intersection of gender and Global South origin, and whether rhetorical choices (hedging, contribution framing) mediate the gender tax.

Primary Research Question: Conditional on equivalent AI quality scores at the working paper stage, do female-authored papers undergo larger quality improvements

during peer review, suggesting that women are held to higher standards?

Secondary Research Questions:

1. On which *specific dimensions* of paper quality (empirical strategy, writing style, contribution framing, data presentation) does the gender delta diverge most?
2. Do rhetorical strategies—hedging, defensive writing, contribution framing—differ by gender, and do they mediate the quality improvement gap?
3. Is the gender tax amplified for economists from Global South institutions and non-English-speaking countries (the intersectional penalty)?
4. Does the pattern vary by author seniority, co-authorship composition, or field?

2 Contribution to the Literature

This study makes five contributions:

First, it introduces AI-based holistic evaluation as a methodological tool for studying bias in expert assessment. [Hengel \(2022\)](#) measured writing quality via readability formulas—mechanical functions of sentence length and syllable counts that capture surface-level text properties. [Alexander et al. \(2023\)](#) used citations as a quality proxy, acknowledging its limitations (endogeneity to peer review, non-classical measurement error). Our AI evaluator assesses *twelve distinct dimensions* of paper quality simultaneously, providing a far richer quality control than any single formula or proxy. [Table 2](#) summarizes the comparison.

Table 2: Comparison with Foundational Studies

Dimension	Hengel (2022)	Alexander et al. (2023)	This Paper
Quality measure	5 readability formulas	Forward citations	12 AI sub-scores
Text analyzed	Abstracts only	None (admin data)	Full papers
Dependent variable	Review duration, readability change	Review duration, referee recs	Δ AI Score (12 dimensions)
Data required	NBER-published pairs + review dates	Journal admin data	NBER-published pairs only
Sample	Top-4 journals; 1,988 pairs	1 journal (detailed) + 32 journals	Top-40 journals; 2,000+ pairs
Geographic scope	US/Europe only	US/Europe only	Global, including Global South
Gender-blind benchmark	None	Citations (endogenous)	AI scores (exogenous)

Notes: This table provides a comparative summary of the methodology, data requirements, and sample characteristics between this study and the foundational literature on gender-based differences in the academic review process. Δ AI Score refers to the change in AI-evaluated quality scores between working paper and published versions.

Second, it provides a feasible empirical strategy that does not require review outcome data. Prior studies needed review duration, number of rounds, or referee recommendations—data that almost no journal makes publicly available. Our delta-score design requires only matched working paper–published pairs, which are available at scale from NBER, SSRN, and other repositories.

Third, it moves the analysis of gendered writing standards from surface-level readability to **deep rhetorical and argumentative structure**. By measuring hedging intensity, contribution framing strength, argumentative aggressiveness, and defensive writing patterns, we test whether the “self-censoring hypothesis” (Hengel, 2022) extends beyond readability to the persuasive fabric of academic papers. This connects the economics literature with the linguistics literature on gendered academic discourse (Hyland, 2005; Biber and Gray, 2016).

Fourth, it provides the first analysis of the **intersection of gender and geographic origin** in academic peer review, testing whether the gender tax documented by Hengel (2022) and Alexander et al. (2023) is amplified for researchers from the Global South. This addresses a critical gap: the entire existing literature focuses on authors at Western institutions.

Fifth, it scales the analysis far beyond the top-four journals studied by Hengel (2022) and the single field journal studied by Alexander et al. (2023), covering dozens of economics journals and including authors from diverse institutional and geographic backgrounds.

3 Data

3.1 Paper Corpus

We construct a dataset of matched working paper–published paper pairs from:

- **NBER Working Papers** (1990–2025): Matched to published versions using the strategy in Hengel (2022). Both the working paper and the final published PDF are obtained for AI evaluation. Estimated N : 2,000–4,000 matched pairs (after power analysis; see Section 5).
- **SSRN Economics Research Network**: Working papers with metadata. Used for the extended baseline analysis (working papers that were never published provide a selection-free comparison).
- **IZA Discussion Papers and World Bank Policy Research Working Papers**: Provide matched pairs from institutions with stronger Global South representation than NBER.

- **OpenAlex / Semantic Scholar APIs:** For bibliometric metadata (citations, references, field classifications, author affiliations, open-access full texts).

3.2 Author Classification

Gender. Following Hengel (2022) and Alexander et al. (2023): (1) name-based classification (GenderAPI, Genderize.io, WIPO gender database), with country-specific name lists for non-Western names; (2) manual verification for ambiguous cases using faculty profiles and institutional pages; (3) paper gender defined as the proportion of female authors, with papers classified as female-authored if the ratio is ≥ 0.50 .

Geographic origin. We classify authors along two dimensions:

- **Institutional affiliation:** Country and institution of the author at time of publication, classified as Core (US, UK, Canada, Australia, Western Europe), Semi-periphery (Southern/Eastern Europe, Israel, Japan, South Korea, Singapore), or Global South (Latin America, Sub-Saharan Africa, South Asia, Southeast Asia, Middle East/North Africa).
- **English nativeness:** Classified by country of affiliation (native English-speaking countries vs. others), refined by author bio and institutional information where available.

3.3 AI Quality Evaluation (ResearchAI)

Each paper—both the working paper version and the published version—is processed through the ResearchAI multi-agent system, which produces:

- **12 section-specific scores** (0–100) from specialist AI agents: Abstract, Introduction, Literature Review, Data, Empirical Strategy, Results, Discussion, Conclusion, Technical Correctness, Writing Style, Tables/Figures, and a Holistic Editorial assessment.
- **Composite score:** Holistic (50%) + Core Section Average (30%) + Minor Section Average (20%).

The AI evaluator receives **only the paper text**—all author names, affiliations, acknowledgments, and publication venue information are stripped before evaluation.

3.4 NLP-Extracted Rhetorical Features

For each paper, we extract rhetorical features using a combination of dictionary-based NLP and LLM-assisted classification:

- **Hedging intensity:** Density of epistemic hedges (“may,” “might,” “suggests,” “appears to”) and scope hedges (“in this context,” “under these assumptions”) per sentence and per section, using established academic hedging lexicons (Hyland, 2005).
- **Boosting intensity:** Density of boosters (“clearly,” “obviously,” “demonstrate,” “prove”). We compute the hedge-to-boost ratio as a summary measure.
- **Contribution framing strength:** Classification of introduction claims as strong (“we are the first to...”), moderate (“we contribute to...”), or weak (“we add to an existing literature...”). Count of distinct contributions listed.
- **Causal language intensity:** Classification of result statements as causal (“X causes Y”), suggestive (“X is associated with Y”), or hedged (“X may affect Y”).
- **Defensive writing thickness:** Ratio of robustness checks/tables to main results; length of identification strategy defense relative to paper length.
- **Self-citation rate:** Proportion of references to own prior work.

3.5 Readability Measures

Following Hengel (2022), we compute Flesch Reading Ease, Flesch-Kincaid, Gunning fog, SMOG, and Dale-Chall scores for all papers.

4 Empirical Strategy

4.1 Step 1: Baseline Balance Test (Working Papers)

We first test whether AI scores for working papers differ by gender:

$$\text{AIScore}_{jk,WP} = \alpha_k + \beta_{1k} \text{female_ratio}_j + \Theta_k \mathbf{X}_j + \varepsilon_{jk} \quad (1)$$

where:

$\text{AIScore}_{jk,WP}$ is the AI score on dimension k (e.g., writing style, empirical strategy, introduction quality) for the *working paper version* of paper j . We estimate this equation separately for each of the 12 sub-scores and for the composite score.

female_ratio_j is the proportion of female authors on paper j , defined as in Hengel (2022). For solo-authored papers, this equals 1 (female) or 0 (male).

$\mathbf{X}_j$ is a vector of controls including: JEL field fixed effects, year fixed effects, number of co-authors, paper length (word count), and the working paper repository (NBER, IZA, World Bank, SSRN).

β_{1k} is the coefficient of interest: the gender gap in AI-assessed quality on dimension k at the working paper stage.

Hypothesis 1 (Quality Equivalence): If $\beta_{1k} \approx 0$ across all dimensions k , then male- and female-authored working papers start at equivalent AI-assessed quality. The playing field is level at baseline, and any differences in the published versions must arise during peer review.

Hypothesis 1a (Writing Quality Premium): If the AI writing style sub-score shows $\beta_{1,\text{writing}} > 0$, women’s working papers are already better written before review— independently confirming Hengel’s readability finding using a fundamentally different methodology.

We run this specification separately for **solo-authored papers** (cleanest gender signal, no co-author confounding) and for all papers.

4.2 Step 2: The Delta Score — Quality Change During Peer Review

For matched working paper–published pairs, we compute the quality change on each dimension:

$$\Delta \text{AIScore}_{jk} = \text{AIScore}_{jk,\text{pub}} - \text{AIScore}_{jk,\text{WP}} \quad (2)$$

where $\Delta \text{AIScore}_{jk}$ is the improvement in AI score on dimension k for paper j between the working paper and published version. A positive delta means quality improved during peer review; a larger delta means more revision was demanded or performed.

We then estimate:

$$\Delta \text{AIScore}_{jk} = \rho_{0k} + \rho_{1k} \text{female_ratio}_j + \rho_{2k} \text{AIScore}_{jk,\text{WP}} + \mathbf{\Lambda}_k \mathbf{X}_j + \varepsilon_{jk} \quad (3)$$

where:

$\Delta \text{AIScore}_{jk}$ is the quality improvement on dimension k (the dependent variable).

female_ratio_j is the proportion of female authors.

$\text{AIScore}_{jk,\text{WP}}$ is the baseline (working paper) quality on dimension k , included to control for mean reversion—papers that start lower tend to improve more mechanically.

$\mathbf{X}_j$ includes journal fixed effects (the journal where the paper was published), year fixed effects, JEL codes, number of co-authors, and author seniority (years since first publication).

ρ_{1k} is the coefficient of interest: the *additional quality improvement* on dimension k that female-authored papers undergo during peer review, conditional on starting quality.

Hypothesis 2 (Gender Tax): If $\rho_{1k} > 0$ for specific dimensions (e.g., writing style, contribution framing), peer review demands more improvement from women on those dimensions—direct evidence of higher standards.

This extends Hengel’s readability-change analysis to all twelve quality dimensions simultaneously. The key advantage: **we do not need any review outcome data** (duration, rounds, referee reports). The AI score delta *is* the outcome.

4.3 Step 3: Published Quality Conditional on Baseline

We test whether, conditional on the same working paper quality, published papers by women end up at a higher quality bar:

$$\text{AIScore}_{jk,\text{pub}} = \gamma_{0k} + \gamma_{1k} \text{female_ratio}_j + \gamma_{2k} \text{AIScore}_{jk,\text{WP}} + \gamma_{3k} (\text{female_ratio}_j \times \text{AIScore}_{jk,\text{WP}}) + \Phi_k \mathbf{X}_j + \varepsilon_j \quad (4)$$

where:

$\text{AIScore}_{jk,\text{pub}}$ is the published version’s AI score on dimension k (the dependent variable).

$\text{AIScore}_{jk,\text{WP}}$ is the working paper version’s score on the same dimension (the baseline quality control).

γ_{1k} captures the gender gap in published quality *conditional on starting quality*. If $\gamma_{1k} > 0$, women’s published papers score higher than men’s conditional on the same WP score—consistent with peer review pushing women further.

γ_{3k} captures whether the gender gap varies across the quality distribution. If $\gamma_{3k} > 0$, the gender tax is larger for higher-quality working papers (high-quality women face even more scrutiny).

4.4 Step 4: Rhetorical Mediators

We test whether gender differences in rhetorical strategy mediate the quality improvement gap:

$$R_{kj} = \alpha_k + \beta_{1k} \text{female_ratio}_j + \beta_{2k} \text{AIScore}_{j,\text{composite}} + \Theta_k \mathbf{X}_j + \varepsilon_{kj} \quad (5)$$

where:

R_{kj} is a rhetorical feature of paper j : hedging intensity, boosting intensity, contribution framing strength, causal language intensity, defensive writing thickness, or self-citation rate.

female_ratio_j is the proportion of female authors.

$\text{AIScore}_{j,\text{composite}}$ is the composite AI quality score, included so that β_{1k} captures the gender gap in rhetoric *conditional on overall quality*—not simply a reflection of quality differences.

β_{1k} is the gender gap in rhetorical feature k .

Hypotheses: If women face higher implicit standards: (H3a) $\beta_{1,\text{hedging}} > 0$ —women hedge more; (H3b) $\beta_{1,\text{framing}} < 0$ —women frame contributions less aggressively; (H3c) $\beta_{1,\text{defensive}} > 0$ —women write more robustness checks; (H3d) $\beta_{1,\text{self-cite}} < 0$ —women self-cite less.

For the matched sample, we estimate the **rhetorical change** during review:

$$\Delta R_{kj} = R_{kj,\text{pub}} - R_{kj,\text{WP}} = \phi_{0k} + \phi_{1k} \text{female_ratio}_j + \Lambda_k \mathbf{X}_j + \varepsilon_{kj} \quad (6)$$

where ΔR_{kj} is the change in rhetorical feature k between working paper and published version. If $\phi_{1,\text{hedging}} > 0$, peer review *demands more hedging from women*. If $\phi_{1,\text{framing}} < 0$, reviewers *force women to weaken their claims more* than men.

4.5 Step 5: The Intersectional Penalty (Gender $\times$ Global South)

We test whether the gender tax compounds at the intersection with geographic origin:

$$\Delta \text{AIScore}_{jk} = \eta_{0k} + \eta_{1k} \text{female}_j + \eta_{2k} \text{GS}_j + \eta_{3k} (\text{female}_j \times \text{GS}_j) + \eta_{4k} \text{AIScore}_{jk,\text{WP}} + \Omega_k \mathbf{X}_j + \varepsilon_{jk} \quad (7)$$

where:

$\Delta \text{AIScore}_{jk}$ is the quality improvement on dimension k during peer review (same dependent variable as Equation 3).

female_j is an indicator for whether the paper is female-authored ($\text{female_ratio} \geq 0.50$).

GS_j is an indicator for whether the corresponding author's affiliation is in the Global South (Latin America, Sub-Saharan Africa, South Asia, Southeast Asia, MENA).

η_{1k} captures the gender tax for Core-institution authors (the Hengel/Alexander baseline).

η_{2k} captures the geographic penalty for male Global South authors.

η_{3k} is the **intersectional penalty**: the *additional* quality improvement demanded from female Global South authors, *beyond* the sum of the gender and geography penalties separately. If $\eta_{3k} > 0$, the penalties compound super-additively.

$AIScore_{jk,WP}$ controls for baseline quality (mean reversion).

$\mathbf{X}_j$ includes journal, year, JEL, co-author count, institutional rank, English nativeness indicator, and author seniority.

We extend the intersectional analysis with the English fluency channel:

$$\Delta AIScore_{jk} = \mu_{0k} + \mu_{1k} \text{female}_j + \mu_{2k} \text{NonNative}_j + \mu_{3k} (\text{female}_j \times \text{NonNative}_j) + \mu_{4k} \text{Readability}_{j,WP} + \Psi_k \mathbf{X}_j \quad (8)$$

where:

NonNative_j indicates whether the author is from a non-native English-speaking country.

$\text{Readability}_{j,WP}$ is the Flesch Reading Ease score of the working paper, controlling directly for the writing quality channel.

μ_{3k} captures the **double writing standard**: the additional revision demanded from female non-native English speakers, beyond the separate effects of gender and language background.

If μ_{3k} falls substantially after controlling for readability, the intersectional penalty operates partly through writing quality. If μ_{3k} persists, it reflects differential treatment beyond writing quality.

4.6 Step 6: Decomposition

We decompose the intersectional penalty into explained (quality) and unexplained (treatment) components using an Oaxaca-Blinder framework:

$$\underbrace{\bar{Y}_{MW} - \bar{Y}_{FGS}}_{\text{Raw gap}} = \underbrace{\sum_{k=1}^{12} \hat{\phi}_k (\bar{S}_{k,MW} - \bar{S}_{k,FGS})}_{\text{Explained: AI quality differences}} + \underbrace{\hat{\eta}_1 + \hat{\eta}_2 + \hat{\eta}_3}_{\text{Unexplained: differential treatment}} \quad (9)$$

where:

$\bar{Y}_{MW}$ is the mean peer review outcome for male authors at Western (Core) institutions—the reference group.

$\bar{Y}_{FGS}$ is the mean outcome for female authors at Global South institutions—the intersectional group.

$\bar{S}_{k,MW}, \bar{S}_{k,FGS}$ are mean AI sub-scores on dimension k for the reference group (male, Western) and the intersectional group (female, Global South), respectively.

$\hat{\phi}_k$ are the estimated returns to quality on dimension k (from the reference group’s regression).

- The **explained component** measures how much of the raw gap is attributable to genuine quality differences on each of the 12 AI dimensions.
- The **unexplained component** measures the residual gap—the portion due to differential treatment by peer reviewers.

This decomposition is novel: no prior study has used AI-predicted quality sub-scores as the “explained” variables in an Oaxaca-Blinder decomposition of peer review outcomes.

4.7 Step 7: Career Trajectory Analysis

Following Hengel’s author-level panel analysis, we test whether rhetorical gender gaps evolve over the career:

$$R_{kijt} = \mu_i + \lambda_t + \gamma_{1k} \text{female}_i \times t_{it} + \gamma_{2k} \text{female}_i \times t_{it}^2 + \Phi_k \mathbf{X}_{ijt} + \varepsilon_{kijt} \quad (10)$$

where:

R_{kijt} is the rhetorical feature k (hedging, framing, etc.) of paper j by author i at career stage t .

μ_i are author fixed effects, absorbing all time-invariant author characteristics (innate writing ability, personality, training).

λ_t are year fixed effects, absorbing trends in academic writing conventions.

t_{it} is the author’s career stage (publication order or years since first publication).

γ_{1k} captures whether the gender gap in rhetorical feature k widens or narrows with experience. If women learn to adapt to higher standards (as Hengel (2022) finds for readability), $\gamma_{1,\text{hedging}}$ should be positive (women hedge more as careers progress) and $\gamma_{1,\text{framing}}$ should be negative (women frame less aggressively over time).

5 Power Analysis and Sample Size Determination

Rather than downloading all available papers, we determine the required sample size through a formal power analysis calibrated to existing effect sizes.

5.1 Effect Size Calibration

Hengel (2022) finds that women’s readability improves 2–5% more during review than men’s. Translating to AI scores: if we assume a similar proportional effect on the writing style sub-score (which operates on a 0–100 scale), this corresponds to approximately 1.5–4 points. With an estimated standard deviation of AI score changes of $\sigma_{\Delta} \approx 12$ –15 points (to be refined by pilot), this implies an effect size of $\delta \approx 0.10$ –0.30 standard deviations.

5.2 Sample Size Calculations

For a two-sample t -test of the gender coefficient in Equation 3, with $\alpha = 0.05$ and power = 0.80:

$$N = \frac{2(z_{\alpha/2} + z_{\beta})^2 \sigma_{\Delta}^2}{\delta^2} \quad (11)$$

where:

N is the total sample size (matched pairs) needed.

$z_{\alpha/2} = 1.96$ is the critical value for $\alpha = 0.05$ (two-sided).

$z_{\beta} = 0.84$ is the critical value for 80% power.

σ_{Δ} is the standard deviation of Δ AI Score.

δ is the minimum detectable effect size.

Table 3: Required Sample Size by Effect Size (Single Dimension)

Effect Size (δ in SD)	N per group	Total N
0.25	252	504
0.20	394	788
0.15	700	1,400
0.10	1,571	3,142

5.3 Multiple Testing Adjustment

With 12 sub-scores, applying a Bonferroni correction ($\alpha_{\text{adj}} = 0.05/12 \approx 0.004$) increases the required sample size by a factor of approximately 1.7. For a target effect of $\delta = 0.20$ SD with Bonferroni: $N \approx 1,340$ total pairs. For the intersectional analysis (four cells: Male $\times$ Core, Male $\times$ GS, Female $\times$ Core, Female $\times$ GS), the smallest cell determines power; target $N \geq 200$ per cell $\times 4 = 800$ minimum.

5.4 Pilot Protocol

Before the full data collection:

1. Obtain 100 NBER WP–published matched pairs (50 male solo-authored, 50 female solo-authored).
2. Run both versions through ResearchAI to estimate σ_{Δ} and the preliminary gender gap $\hat{\rho}_1$.
3. Use pilot estimates to refine sample size and confirm feasibility.
4. Target final sample: **2,000–3,000 matched pairs**, stratified to oversample female authors and Global South authors.

6 Identification Challenges and Robustness

AI Bias. If the AI evaluator itself embeds gender biases, our baseline balance test (Equation 1) is compromised. We address this by: (a) stripping all author information before evaluation; (b) running the audit study in Agenda 2 *before* deploying the AI counterfactual in Agenda 1; (c) comparing AI scores with readability formulas (which are algorithmic and bias-free by construction).

Selection Bias. Published papers are a selected sample. We mitigate this by: (a) including never-published working papers for the baseline analysis; (b) exploiting within-paper variation (pre- vs. post-review); (c) comparing AI score distributions across journals of different selectivity.

AI Score Calibration. LLM-based scores may suffer from scale compression. We address this by: (a) using rank-based measures alongside raw scores; (b) standardizing scores within journal-year cells; (c) examining sensitivity to prompt engineering and model choice.

Gender Assignment for Non-Western Names. Name-based gender prediction is less accurate for South Asian, East Asian, and Middle Eastern names. We invest in manual verification for the intersectional sample and report robustness restricting to manually verified gender assignments.

Mean Reversion. Lower-quality working papers tend to improve more mechanically. We control for baseline AI score ($\text{AIScore}_{jk,WP}$) in all delta regressions.

7 Validation: AI vs. Human Evaluation

To validate the AI evaluator, we test:

1. Whether AI scores predict publication venue quality (top-5 > field journals > unpublished working papers).
2. Whether AI scores predict subsequent citations (3- and 5-year forward citations).
3. Whether AI writing style sub-scores correlate with Flesch/Gunning readability measures (convergent validity with Hengel 2022).
4. Whether AI scores improve from WP to published version on average (face validity: peer review should improve papers).

8 Expected Results and Policy Implications

If the gender tax persists across multiple AI quality dimensions—women’s papers improve more during peer review conditional on starting quality—this provides the strongest evidence to date that peer review systematically holds women to higher standards.

The dimension-specific analysis reveals *where* to intervene: if the penalty is concentrated in “contribution framing” or “writing style,” journals could develop structured evaluation rubrics for these dimensions. The rhetorical analysis reveals *how* women adapt: if women hedge more and frame contributions less aggressively over their careers, this quantifies the cost of self-censoring.

The intersectional analysis reveals *who* bears the largest burden: if the gender tax compounds super-additively with Global South origin, then policy interventions (AI writing tools, structured review, referee training) should target the intersection, not each dimension separately.

9 Literature Review Strategy

9.1 The Five Literature Strands

9.1.1 Strand 1: Gender Bias in Peer Review

Key papers: Hengel (2022), Alexander et al. (2023), Card et al. (2020), Hengel and Moon (2022), Sarsons et al. (2021). *Your positioning:* you replace unidimensional quality

measures with 12-dimensional AI assessment, do not require review outcome data, and extend to Global South authors.

9.1.2 Strand 2: AI and Algorithmic Evaluation in Expert Settings

Key papers: Kleinberg et al. (2018) (AI as counterfactual in bail decisions), Cowgill (2019, AI in hiring), Checco et al. (2021) (AI-assisted peer review). *Your positioning:* first application of AI-as-counterfactual to academic peer review.

9.1.3 Strand 3: Gendered Academic Discourse and Rhetoric

Key papers: Hyland (2005) (hedging/boosting), Biber and Gray (2016) (grammatical complexity), Gentzkow et al. (2019) (text-as-data in economics), Exley and Kessler (2022) and Coffman (2014) (self-promotion and self-censoring). *Your positioning:* first study to link NLP-measured rhetorical features to peer review outcomes by gender in economics.

9.1.4 Strand 4: Intersectionality in Economics

Key papers: Bayer and Charles (2018) (intersectional analysis methodology), Crenshaw (1989, foundational concept), Das et al. (2013) (geography of academic research), Greenman and Xie (2008, double jeopardy). *Your positioning:* first intersectional analysis of gender × geography in peer review.

9.1.5 Strand 5: Discrimination and Information

Key papers: Bohren et al. (2019) (dynamic discrimination), Brock and De Haas (2022) (evaluator experience and bias), Phelps (1972, statistical discrimination). *Your positioning:* referees “novice” in familiarity with Global South institutions behave like novice referees in Alexander et al.’s framework.

9.2 Critical Literature Gap: The 2×2 Matrix

		Gender bias studied	Gender bias not studied
Geography studied	Your paper (first!)		Das et al. 2013
Geography not studied	Hengel 2022; Alexander et al. 2023		—

10 Timeline

Table 4: Timeline for Agenda 1

Phase	Activity	Timeline
1	Pilot study (100 matched pairs)	Months 1–2
2	Power analysis refinement and sample size determination	Month 2
3	Web scraping pipeline (NBER, IZA, WB, SSRN, OpenAlex)	Months 2–5
4	Author classification (gender, geography, English nativeness)	Months 3–6
5	AI evaluation of matched pairs (batch processing)	Months 5–8
6	NLP pipeline: rhetorical feature extraction	Months 5–7
7	Econometric analysis (Steps 1–7)	Months 8–12
8	Robustness, validation, and decomposition	Months 11–14
9	Paper writing and submission	Months 13–17

Research Agenda 2

Does the AI Referee Discriminate? Auditing Gender and Geographic Bias in Large Language Model Evaluations of Academic Economics

Target Journals: *American Economic Review: Insights, Nature, Proceedings of the National Academy of Sciences*

11 Motivation and Research Question

Large language models are rapidly being adopted as tools for academic evaluation: manuscript screening, grant review, teaching evaluation, and hiring shortlisting. If these tools replicate the gender biases documented in human peer review (Hengel, 2022; Alexander et al., 2023; Card et al., 2020), their deployment could *institutionalize* discrimination at unprecedented scale. Conversely, if LLMs are genuinely identity-blind, they offer a powerful debiasing mechanism.

This paper provides the first rigorous, large-scale audit of **both gender and geographic bias** in LLM-based academic evaluation. We send identical papers through multiple AI evaluation systems under experimentally varied conditions: with author identity revealed vs. concealed, with male vs. female names swapped, and with Western vs. Global South institutional affiliations manipulated. The multi-identity design—simultaneously varying gender, ethnicity, and geography—goes beyond existing correspondence studies, which typically vary only one dimension.

This paper also serves a critical **validation function** for Agenda 1: before using AI scores as a gender-blind benchmark for studying human bias, we must verify that the AI itself does not discriminate.

Primary Research Question: Do large language models exhibit gender or geographic bias when evaluating academic economics papers, and if so, does the bias mirror the patterns found in human peer review?

Secondary Research Questions:

1. Does revealing author gender to the AI (through names) alter its evaluation, and on which quality dimensions?

2. Does revealing Global South institutional affiliations or non-Western names produce lower scores, even controlling for paper quality?
3. Do AI models replicate Hengel’s finding that female-authored papers are held to higher writing standards?
4. How do bias patterns vary across AI models (Gemini, GPT-4, Claude, Llama)?
5. If bias exists, can it be mitigated through blinding, prompt engineering, structured rubrics, or multi-model ensembles?

12 Contribution to the Literature

First, this paper provides the first **audit study** of gender and geographic bias in AI-based academic evaluation. The AI fairness literature has studied bias in hiring (Raghavan et al., 2020), criminal justice (Dressel and Farid, 2018), and healthcare (Obermeyer et al., 2019), but not in academic peer review. Given the rapid adoption of AI in academic settings (Checco et al., 2021), this is an urgent gap.

Second, we extend the **correspondence study methodology** to AI evaluators with a multi-identity design. Traditional correspondence studies (Bertrand and Mullainathan, 2004; Kline et al., 2022) send identical applications with different names to human evaluators. We send identical papers with different author identities to AI evaluators, achieving **perfect experimental control**: the same paper can be evaluated multiple times under different identity conditions, eliminating evaluator heterogeneity—something impossible with human reviewers.

Third, we go beyond the single-dimension designs typical of correspondence studies by simultaneously varying **gender, ethnicity, and institutional affiliation**. This allows us to test for intersectional bias: does an AI penalize a female author with a non-Western name at a Global South institution more than the sum of each penalty?

Fourth, we validate the identifying assumption underlying Agenda 1: that AI scores are gender-blind when author information is stripped. If confirmed, this strengthens the entire research program.

13 Experimental Design

13.1 The Three-Layer Audit Framework

Layer 1: Blind Evaluation (Baseline). Papers are stripped of all author information (names, affiliations, acknowledgments, identifiable self-citations). The AI evaluates each paper purely on content. This produces the **identity-blind benchmark**.

Layer 2: Identity-Revealed Evaluation (Audit). The same papers are evaluated with author information visible. We use a full-factorial design:

Table 5: Identity Manipulation Conditions

Dimension	Conditions	Example
Gender	Male / Female / Ambiguous (initials)	John Smith / Jane Smith / J. Smith
Ethnicity	Western / South Asian / East Asian / Hispanic	Smith / Sharma / Chen / García
Affiliation	Top US / Mid-tier US / Global South	MIT / Univ. of Nebraska / Univ. del

Name swaps are matched on frequency and generational cohort. We generate 8–12 identity profiles per paper (male-Western-TopUS, female-Western-TopUS, male-SouthAsian-GlobalSouth, female-SouthAsian-GlobalSouth, etc.) to test main effects and interactions.

Layer 3: Cross-Model Comparison. Each paper-condition pair is evaluated by multiple AI models:

- Google Gemini (ResearchAI’s default)
- OpenAI GPT-4 / GPT-4o
- Anthropic Claude
- Meta Llama (open-source baseline)

This identifies whether bias is model-specific or systematic across LLMs.

13.2 Paper Sample

Sample A: Published papers with known outcomes ($N = 2,000\text{--}5,000$). Drawn from top-40 economics journals (2000–2025), with matched NBER/SSRN working paper versions where available. Gender assigned as in [Hengel \(2022\)](#).

Sample B: Controlled synthetic variants ($N = 500$). Created by systematically manipulating: (a) readability level (high/medium/low); (b) contribution framing strength (strong/moderate/weak); (c) methodology rigor (clean/moderate threats/weak identification). This full-factorial design isolates AI sensitivity to each textual feature independently of author identity.

14 Empirical Strategy

14.1 The Audit Test: Does Author Identity Affect AI Scores?

For each paper j evaluated under identity condition c :

$$\text{AIScore}_{jc} = \alpha + \beta_1 \text{FemaleName}_{jc} + \beta_2 \text{NonWesternName}_{jc} + \beta_3 \text{GS_Affil}_{jc} + \beta_4 (\text{FemaleName}_{jc} \times \text{GS_Affil}_{jc}) \quad (12)$$

where:

AIScore_{jc} is the AI score for paper j under identity condition c (the dependent variable).

FemaleName_{jc} is an indicator equal to 1 if the author name in condition c is female (e.g., “Jane Smith”), 0 otherwise. The blind condition is the omitted category.

$\text{NonWesternName}_{jc}$ is an indicator equal to 1 if the name is non-Western (South Asian, East Asian, Hispanic, etc.).

GS_Affil_{jc} is an indicator equal to 1 if the affiliation listed is a Global South institution.

β_4 captures the **intersectional AI bias**: does the AI penalize female authors from the Global South more than the sum of the gender and geography penalties?

PaperFE_j are paper fixed effects, absorbing *all* paper-level quality. With paper fixed effects, β_1 , β_2 , β_3 , and β_4 are identified purely from within-paper variation in the identity treatment. This is the key advantage over human audit studies.

The primary test is $\hat{\beta}_1$: the causal effect of presenting a female (vs. blind) author name on the *same* paper’s AI score.

We decompose this into sub-scores:

$$\text{SubScore}_{kjc} = \alpha_k + \beta_{1k} \text{FemaleName}_{jc} + \beta_{2k} \text{NonWesternName}_{jc} + \beta_{3k} \text{GS_Affil}_{jc} + \gamma_k \text{PaperFE}_j + \varepsilon_{kjc} \quad (13)$$

where SubScore_{kjc} is the AI score on dimension k (writing style, empirical strategy, introduction, etc.) for paper j under condition c . This identifies *which evaluation dimensions* are most susceptible to identity-based bias.

14.2 Does AI Bias Mirror Human Bias?

We compare AI bias patterns with documented human bias:

Test 1 (Hengel replication): Among papers evaluated blindly, do female-authored papers receive higher writing style scores? If yes, the AI independently confirms that women write better.

Test 2 (Alexander et al. replication): When author gender is revealed, does the AI apply higher thresholds for female-authored papers? We test whether score thresholds for implicit “accept” recommendations shift by gender.

Test 3 (Agent asymmetry): Do different AI agents (the Holistic Editor vs. specialist reviewers) show different bias patterns, paralleling Card et al.’s finding that editors and referees behave differently?

14.3 Implicit Gender Detection Test

Before the audit, we test whether the AI can detect author gender from *blinded text alone*:

$$P(\text{AI guesses female} \mid \text{paper text}_j) = \Lambda(\alpha + \beta \text{TextFeatures}_j) \quad (14)$$

where:

$\Lambda(\cdot)$ is the logistic function.

TextFeatures_j are NLP-extracted features (hedging density, self-citation rate, contribution framing strength, etc.).

- We prompt the AI: “Based only on the writing style and content, what is your best guess of the author’s gender?”

If the AI can detect gender at above-chance rates from blinded text, then “blinding” does not fully eliminate identity information. This would be a powerful finding with implications for both AI evaluation and double-blind review.

14.4 Text Analysis of AI Reviews

We apply NLP to the AI-generated review text to detect subtle bias in *how* the AI discusses male- vs. female-authored work:

- **Tone:** Sentiment, constructiveness, and formality of feedback.
- **Specificity:** Are suggestions more actionable for one gender?
- **Hedging of praise:** “This is a strong contribution” vs. “This appears to be a reasonable paper” for similar scores.
- **Focus allocation:** Does the AI spend more words on methodology for one gender and more on writing for the other?

14.5 Debiasing Experiments

For any detected bias, we test five mitigation strategies:

1. **Blinding:** Simply removing author names (Layer 1). Quantify the bias reduction as a share of the name-revealed bias.
2. **Explicit debiasing instructions:** “Evaluate this paper without regard to author identity. Focus exclusively on research quality.” Test whether stating the goal reduces bias.
3. **Structured evaluation rubrics:** Replace open-ended evaluation with strict rubrics specifying exactly what to assess on each dimension. Rubric-based evaluation constrains AI discretion.
4. **Multi-model ensemble:** Average scores across all four AI models. If different models have different or opposing biases, ensembling may cancel them.
5. **Calibration anchors:** Provide explicit scoring benchmarks (“A score of 75 means. . .”). Test whether anchoring reduces gender-differential scoring.

For each strategy s , we estimate:

$$\text{AIScore}_{jcs} = \alpha_s + \beta_{1s} \text{FemaleName}_{jc} + \gamma_s \text{PaperFE}_j + \varepsilon_{jcs} \quad (15)$$

where:

s indexes the debiasing strategy (including no-strategy baseline).

β_{1s} is the gender bias under strategy s .

- The effectiveness of strategy s is measured as $1 - |\hat{\beta}_{1s}|/|\hat{\beta}_{1,\text{baseline}}|$: the proportional reduction in bias.

15 Connection to the Research Program

15.1 The AI as Benchmark for Agenda 1

If AI evaluators show **no bias when blinded** ($\hat{\beta}_1 \approx 0$ in Layer 1):

- AI scores provide a clean quality benchmark for Agenda 1.
- Supports using AI tools in journal pre-screening to reduce human bias.
- Validates Hengel’s assumption that quality differences do not explain the gap.

If AI evaluators show **bias even when blinded** (e.g., through stylistic cues correlated with gender):

- Raises concerns about using AI for academic evaluation.
- Suggests that “gender-blind” evaluation may be impossible if gendered writing styles are detectable.
- Requires adjusting Agenda 1’s methodology to account for AI bias.

15.2 Calibrating Human vs. AI Bias

Table 6: Scenario Matrix: AI Bias and Policy Implications

AI Blind Bias	AI Revealed Bias	Name-	Implication
None	None		AI is identity-neutral; deploy widely for pre-screening
None	Present		AI responds to identity cues; use only with blinding
Present (pro-female)	Present (anti-female)	(anti-	Biases partially cancel; ensemble with blinding optimal
Present (anti-female)	Amplified		AI replicates human bias; do NOT deploy without mitigation

16 Literature Review Strategy

16.1 The Five Literature Strands

16.1.1 Strand 1: AI Fairness and Algorithmic Bias (Primary)

Key papers: Obermeyer et al. (2019) (*Science*—gold standard for algorithmic bias audit, the paper referees will compare you to); Dressel and Farid (2018) (algorithms and humans have similar bias); Raghavan et al. (2020) (bias in algorithmic hiring); Liang et al. (2023) (GPT detectors biased against non-native English writers). *Your positioning:* extend the AI fairness literature to academic peer review, with a multi-identity (gender $\times$ ethnicity $\times$ geography) design.

16.1.2 Strand 2: Correspondence Study Methodology

Key papers: [Bertrand and Mullainathan \(2004\)](#) (canonical audit study); [Kline et al. \(2022\)](#) (modern large-scale audit). *Your advantage:* perfect experimental control—same paper, same AI, different identities.

16.1.3 Strand 3: Gender Bias in Peer Review

Key papers: [Hengel \(2022\)](#), [Alexander et al. \(2023\)](#), [Card et al. \(2020\)](#). Frame each as posing a question the AI might replicate.

16.1.4 Strand 4: AI Adoption in Academia (Urgency)

Search for journal editorial policies on AI in peer review as of 2025–2026. Cite adoption evidence.

16.1.5 Strand 5: Debiasing Interventions for AI

Key references: [Schick et al. \(2021\)](#) on self-debiasing; [Barocas, Hardt, and Narayanan \(2023\)](#) *Fairness and Machine Learning*.

16.2 Critical Week-1 Confirmation Search

Search for: “audit study” AND “LLM” AND “academic evaluation” AND “gender”; “correspondence study” AND “AI” AND “peer review”; “gender bias” AND “AI referee.” If < 5 results exist as of 2026, your contribution is clear.

17 Timeline

Table 7: Timeline for Agenda 2

Phase	Activity	Timeline
1	Pilot: 20 papers $\times$ 4 identity conditions $\times$ 2 models	Months 1–2
2	Sample construction, name/affiliation swapping	Months 2–4
3	Layer 1 evaluations (blind, all models)	Months 3–5
4	Layer 2 evaluations (identity-revealed, all conditions)	Months 4–7
5	Implicit gender detection experiment	Month 5
6	Debiasing experiments	Months 6–9
7	NLP analysis of AI review text	Months 7–10
8	Statistical analysis and validation	Months 9–12
9	Paper writing and submission	Months 11–15

Research Agenda 3

Did AI Tools Close the Gender Gap? Evidence from the Generative AI Revolution

Target Journals: *Quarterly Journal of Economics*, *American Economic Review*, *Science*

18 Motivation and Research Question

Hengel (2022) documents a vicious cycle: female economists face higher writing standards in peer review, which forces them to spend more time polishing manuscripts, leaving less time for new research. Alexander et al. (2023) show that this cycle costs women 29–60 extra days per paper at *Energy Economics* alone. Both papers identify the problem but offer limited solutions.

On November 30, 2022, OpenAI released ChatGPT. On March 14, 2023, GPT-4 followed. These events constitute the largest exogenous shock to academic writing technology in modern history. For the first time, all researchers—regardless of gender, seniority, institutional resources, or native language—gained access to powerful AI writing and editing assistance. If the gender gap documented by Hengel (2022) is partly driven by **supply-side** factors (women must invest more effort to meet higher writing standards), then the sudden availability of AI tools should *disproportionately benefit women* by reducing the effort cost of quality equalization. If the gap is primarily **demand-side** (reviewers discriminate regardless of quality), the AI revolution should have no differential effect by gender.

This paper exploits the generative AI revolution as a **natural experiment** to answer the same questions an RCT would, without the need for participant recruitment, IRB approval, or multi-year outcome tracking. We compare the gender gap in AI-assessed paper quality and in the quality delta (published minus working paper) for papers written *before* vs. *after* AI tools became available, using the same data infrastructure as Agenda 1.

Primary Research Question: Did the availability of generative AI tools reduce the gender gap in working paper quality at submission, and did it narrow the quality improvement demanded of women during peer review?

Secondary Research Questions:

1. On which quality dimensions (writing style, contribution framing, empirical strategy) did AI availability have the largest equalizing effect?
2. Did the narrowing of the gender gap (if any) differ between Core and Global South institutions, consistent with differential AI adoption rates?
3. Did AI availability affect the *rhetorical* gender gap—did women’s hedging, contribution framing, and defensive writing converge toward male patterns after AI tools became available?
4. Can we distinguish supply-side equalization (better manuscripts at submission) from demand-side persistence (reviewers still demand more from women)?
5. Did AI tools narrow or widen the geographic quality gap between Core and Global South working papers?

19 Contribution to the Literature

First, this paper provides the first quasi-experimental evidence on whether AI writing tools reduce gender disparities in academic output quality. The growing literature on AI as equalizer (Brynjolfsson et al., 2023; Noy and Zhang, 2023) has studied customer service, professional writing, and coding, but not academic research. We extend this literature to the highest-stakes writing domain: academic publishing, where quality disparities have career consequences.

Second, we provide a **supply vs. demand decomposition** of the gender gap without requiring an RCT. If the gender gap in *working paper* quality (supply) narrows post-AI while the gender gap in *deltas* (demand) persists, we can attribute the remaining gap to reviewer treatment. This observational decomposition complements Agenda 1’s cross-sectional analysis with time-series variation.

Third, we test a critical **distributional hypothesis** about AI technology: does it compress or widen the quality distribution across gender, geography, and their intersection? Brynjolfsson et al. (2023) find AI compresses the skill distribution among customer service agents. We test whether the same holds among academic economists—a population with much higher baseline skill and stronger selection.

Fourth, the paper contributes to the **technology adoption and inequality** literature (Acemoglu and Restrepo, 2020) by examining whether a technology that is nominally available to all is equally *beneficial* to all. If AI adoption is lower or less effective among Global South researchers (due to cost, internet access, English proficiency, or institutional norms), the AI revolution may widen geographic gaps even as it narrows gender gaps within Core institutions.

20 The Natural Experiment: Timeline and Identification

20.1 Key Dates

Table 8: Timeline of the Generative AI Shock

Date	Event
November 30, 2022	ChatGPT released (GPT-3.5)
March 14, 2023	GPT-4 released
March 2023	Google Bard (later Gemini) released
July 2023	Claude 2 released (Anthropic)
Late 2023–2024	Widespread academic adoption; journal policies emerge
2024–2025	Specialized AI writing/editing tools proliferate

20.2 Identification Strategy

The key assumption is that the release of generative AI tools was **exogenous to the gender gap in economics peer review**. OpenAI, Google, and Anthropic did not design these tools to address gender disparities in academic writing—they were general-purpose products whose academic applications were a side effect. The timing of their release was driven by the AI industry’s development cycle, not by any academic publishing consideration.

We define three periods:

- **Pre-AI period:** Working papers circulated before November 2022. Authors had no access to LLM-based writing tools during manuscript preparation.
- **Transition period:** Working papers circulated December 2022–June 2023. Some authors may have adopted AI tools; others had not yet. We exclude this period from the main analysis and include it in robustness.
- **Post-AI period:** Working papers circulated after June 2023. AI tools were widely available and increasingly adopted.

21 Data

We use the **same data infrastructure** as Agenda 1, with two additions:

21.1 Working Paper Corpus with Time Variation

- **NBER Working Papers:** All WPs with circulation dates, partitioned into pre-AI (2018–October 2022) and post-AI (July 2023–2025). Estimated N : 6,000–8,000 pre-AI + 3,000–4,000 post-AI.
- **IZA Discussion Papers:** Broader geographic coverage, especially European and Global South authors.
- **World Bank Policy Research Working Papers:** Strong Global South representation.
- **SSRN Economics Research Network:** Large volume, diverse author base.

For papers that were eventually published, we match to the published version to compute the delta (as in Agenda 1).

21.2 AI Adoption Intensity Proxy

We construct a paper-level proxy for AI-assisted writing using three approaches:

1. **Temporal:** Papers circulated after June 2023 are coded as “AI-available.” This is the intent-to-treat measure.
2. **Text-based:** We compute a battery of stylistic markers associated with LLM-generated text—perplexity scores (using a reference language model), burstiness (variation in sentence complexity), and the prevalence of specific LLM-associated phrases (“it is important to note,” “delve into,” “a nuanced understanding”). We construct an **AI-fingerprint score** that estimates the probability that AI tools contributed to the writing. This is the treatment intensity measure.
3. **Survey-based (supplementary):** For a subsample, we survey authors about their AI tool usage during manuscript preparation. This validates the text-based proxy.

22 Empirical Strategy

22.1 Step 1: Difference-in-Differences on Working Paper Quality

We test whether the gender gap in working paper quality narrowed after AI tools became available:

$$\text{AIScore}_{jk,WP} = \alpha_k + \beta_{1k} \text{female}_j + \beta_{2k} \text{Post}_j + \beta_{3k} (\text{female}_j \times \text{Post}_j) + \Theta_k \mathbf{X}_j + \varepsilon_{jk} \quad (16)$$

where:

$\text{AIScore}_{jk,WP}$ is the AI quality score on dimension k for the working paper version of paper j . We estimate separately for each of the 12 sub-scores and the composite score.

female_j is an indicator for whether paper j is female-authored ($\text{female_ratio} \geq 0.50$).

Post_j is an indicator equal to 1 if the working paper was circulated after June 2023 (post-AI period), and 0 if before November 2022 (pre-AI period).

β_{1k} is the pre-AI gender gap in working paper quality on dimension k .

β_{2k} is the overall change in working paper quality post-AI (common to both genders).

β_{3k} is the **key parameter**: the differential change in the gender gap after AI became available. If $\beta_{3k} > 0$ (e.g., on writing style), AI tools narrowed the gender gap on dimension k —women’s working papers improved more (or men’s improved less) after AI became available.

$\mathbf{X}_j$ includes JEL field fixed effects, working paper repository fixed effects (NBER, IZA, WB, SSRN), number of co-authors, author seniority, and calendar month fixed effects (to control for seasonal patterns).

Interpretation: If Hengel’s finding holds (women already write better at baseline) and AI tools help equalize effort costs, we expect β_{3k} to be *negative* on writing style (the gender premium shrinks because men catch up with AI) but possibly *positive* on other dimensions where women previously under-invested (contribution framing, argumentative aggressiveness).

22.2 Step 2: Event Study

To test the parallel trends assumption and examine the timing of effects, we estimate an event study:

$$\text{AIScore}_{jk,WP} = \alpha_k + \sum_{\tau \neq -1} \gamma_{\tau k} (\text{female}_j \times \mathbf{1}[\text{semester}_j = \tau]) + \delta_k \text{female}_j + \Lambda_k \mathbf{X}_j + \text{semester FE} + \varepsilon_{jk} \quad (17)$$

where:

τ indexes semesters relative to the AI shock. $\tau = -1$ (Spring 2022) is the reference period. $\tau = 0$ is Fall 2022 (ChatGPT release). $\tau = 1$ is Spring 2023 (GPT-4 release). $\tau \geq 2$ is the post-AI period.

$\gamma_{\tau k}$ captures the gender gap in AI score on dimension k in semester τ , relative to the reference period.

- **Pre-trends test:** If $\gamma_{\tau k} \approx 0$ for $\tau < 0$ (the pre-AI semesters), the parallel trends assumption holds—the gender gap was stable before the AI shock.
- **Treatment dynamics:** We expect $\gamma_{\tau k}$ to begin moving in the equalizing direction for $\tau \geq 1$, with the effect growing as adoption spreads.

This event study plot is the paper’s “Figure 1”—a visual demonstration of whether the gender quality gap shifted at the moment AI tools became available.

22.3 Step 3: Diff-in-Diff on the Quality Delta

For papers with matched working paper–published versions, we test whether the gender gap in quality *improvement during peer review* changed post-AI:

$$\Delta \text{AIScore}_{jk} = \alpha_k + \rho_{1k} \text{female}_j + \rho_{2k} \text{Post}_j + \rho_{3k} (\text{female}_j \times \text{Post}_j) + \rho_{4k} \text{AIScore}_{jk, \text{WP}} + \Omega_k \mathbf{X}_j + \varepsilon_{jk} \quad (18)$$

where:

$\Delta \text{AIScore}_{jk}$ is the quality improvement on dimension k between working paper and published version (same as Agenda 1’s delta).

ρ_{1k} is the pre-AI gender gap in quality improvement demanded—the Hengel finding (women must improve more).

ρ_{3k} is the **key parameter**: did the additional improvement demanded from women change after AI tools became available?

$\text{AIScore}_{jk, \text{WP}}$ controls for baseline quality (mean reversion).

Supply vs. demand decomposition:

- If $\beta_{3k} \neq 0$ (Step 1) **and** $\rho_{3k} \neq 0$ (Step 3): AI affected *both* supply (WP quality) and demand (delta). The total effect on published paper quality can be decomposed.

- If $\beta_{3k} \neq 0$ but $\rho_{3k} \approx 0$: AI improved supply but reviewers still demand the same from women—**supply-side equalization, demand-side persistence**. The gap is demand-driven.
- If $\beta_{3k} \approx 0$ but $\rho_{3k} \neq 0$: AI did not change WP quality by gender, but reviewers changed behavior post-AI—suggests reviewers are responding to AI availability, not quality changes.
- If both ≈ 0 : AI had no effect on the gender gap. The gap is structural and unresponsive to technology.

22.4 Step 4: Adoption Intensity and Heterogeneity

We replace the binary Post_j with the continuous AI-fingerprint score to estimate dose-response:

$$\text{AIScore}_{jk, \text{WP}} = \alpha_k + \beta_{1k} \text{female}_j + \beta_{2k} \text{AIFingerprint}_j + \beta_{3k} (\text{female}_j \times \text{AIFingerprint}_j) + \Theta_k \mathbf{X}_j + \varepsilon_{jk} \quad (19)$$

where:

AIFingerprint_j is a continuous score (0 to 1) measuring the estimated probability that paper j was written with AI assistance, based on textual markers (perplexity, burstiness, LLM-associated phrases).

β_{3k} captures whether papers with stronger AI fingerprints show a smaller gender gap on dimension k . A positive β_{3k} on writing quality means: the more AI-assisted the paper, the smaller the female writing premium—consistent with AI equalizing writing effort.

Caution: The AI-fingerprint score is endogenous (adoption is a choice). We use it as a descriptive tool, not for causal identification. The diff-in-diff (Step 1) provides the causal estimate; the fingerprint analysis shows which mechanism drives it.

22.5 Step 5: Global South Heterogeneity

We test whether the AI shock had differential effects across geography:

$$\begin{aligned} \text{AIScore}_{jk, \text{WP}} = & \alpha_k + \beta_{1k} \text{female}_j + \beta_{2k} \text{Post}_j + \beta_{3k} \text{GS}_j + \beta_{4k} (\text{female}_j \times \text{Post}_j) \\ & + \beta_{5k} (\text{GS}_j \times \text{Post}_j) + \beta_{6k} (\text{female}_j \times \text{GS}_j \times \text{Post}_j) + \Theta_k \mathbf{X}_j + \varepsilon_{jk} \end{aligned} \quad (20)$$

where:

β_{4k} is the narrowing of the gender gap post-AI for Core-institution authors.

β_{5k} is the change in the geographic gap post-AI for male authors.

β_{6k} is the **triple difference**: did the intersectional gap (female $\times$ Global South) narrow more or less than predicted by the additive effects? If $\beta_{6k} > 0$, AI especially helped female Global South authors. If $\beta_{6k} < 0$, AI benefits bypassed the intersection—technology widened rather than narrowed intersectional inequality.

This is the core test of whether AI serves as an **intersectional equalizer** or an **inequality amplifier**.

22.6 Step 6: Rhetorical Convergence

We test whether AI tools changed the rhetorical gender gap:

$$R_{kj} = \alpha_k + \phi_{1k} \text{female}_j + \phi_{2k} \text{Post}_j + \phi_{3k} (\text{female}_j \times \text{Post}_j) + \mathbf{\Lambda}_k \mathbf{X}_j + \varepsilon_{kj} \quad (21)$$

where:

R_{kj} is a rhetorical feature of paper j : hedging intensity, contribution framing strength, causal language intensity, defensive writing thickness, or self-citation rate.

ϕ_{1k} is the pre-AI rhetorical gender gap.

ϕ_{3k} captures whether the rhetorical gender gap changed post-AI. If $\phi_{3,\text{hedging}} < 0$, women hedge less after AI—their writing converges toward male rhetoric. If $\phi_{3,\text{framing}} > 0$, women frame contributions more aggressively post-AI.

Interpretation: AI writing tools may not only improve surface quality but also change *how* authors argue. If AI encourages more assertive framing (as LLMs tend to generate confident prose), and if women adopt this framing, the self-censoring hypothesis (Hengel, 2022) may be partially resolved by technology.

23 Identification Challenges and Robustness

23.1 Threats to Identification

Threat 1: Confounding trends. The gender gap may have been narrowing for reasons unrelated to AI (changing norms, DEI initiatives, generational shifts). *Mitigation:* The

event study (Step 2) tests for pre-trends. If the gender gap was flat for semesters $\tau < 0$ and shifts only at $\tau \geq 0$, confounding trends are unlikely.

Threat 2: Composition changes. The pool of working paper authors may differ pre- vs. post-AI (e.g., more junior female researchers entering the field). *Mitigation:* Control for author seniority, institutional rank, and JEL field. Estimate within-author regressions for authors who have papers in both periods.

Threat 3: AI-fingerprint endogeneity. Authors who adopt AI tools may differ systematically from those who don't. *Mitigation:* Use the fingerprint score only for descriptive dose-response analysis. The causal estimate comes from the time-based diff-in-diff, which relies on the exogeneity of AI availability (not adoption).

Threat 4: Journal policy changes. Some journals introduced AI-related policies in 2023–2024 (disclosure requirements, restrictions). These policies may affect manuscript preparation differentially by gender. *Mitigation:* Include journal $\times$ year fixed effects. Test robustness excluding journals with AI-specific policies.

23.2 Robustness Checks

1. **Alternative cutoff dates:** Use November 2022 (ChatGPT) or March 2023 (GPT-4) instead of June 2023. Use a donut specification excluding the transition period.
2. **Placebo test:** Run the same diff-in-diff using a fake “shock” at an earlier date (e.g., January 2020). The interaction should be zero.
3. **Within-author analysis:** For prolific authors with papers in both periods, estimate author fixed effects models. This controls for all time-invariant author characteristics.
4. **Solo-authored papers:** Restrict to solo-authored papers for the cleanest gender signal.
5. **Readability convergence:** Use traditional readability formulas (Flesch, Gunning fog) alongside AI scores to test whether surface-level readability also converged post-AI.
6. **AI detection sensitivity:** Test whether results are robust to different AI-fingerprint score thresholds and construction methods.

24 Power Analysis

24.1 Effect Size Calibration

Brynjolfsson et al. (2023) find that AI tools compress the performance distribution by 0.2–0.4 SD for customer service workers. Noy and Zhang (2023) find AI improves writing quality by 0.3–0.5 SD in their RCT, with larger effects for weaker writers. If we assume a conservative compression of 0.15–0.25 SD in the gender quality gap post-AI:

24.2 Sample Size Adequacy

Table 9: Power Analysis for the Diff-in-Diff ($\alpha = 0.05$, Power = 0.80)

Parameter	MDE (SD)	N pre	N post
Gender $\times$ Post (β_3 , main DID)	0.12	4,000	2,000
Gender $\times$ Post $\times$ GS (β_6 , triple diff)	0.25	4,000	2,000
Gender $\times$ Post on delta (ρ_3 , matched pairs)	0.18	1,500	800

With 6,000–8,000 pre-AI and 3,000–4,000 post-AI working papers from NBER, IZA, WB, and SSRN, the main diff-in-diff is well-powered to detect effects as small as 0.12 SD. The triple difference (intersectional effect) requires larger effects but is feasible with the combined sample.

25 Expected Results and Scenarios

Scenario A: AI Closed the Supply-Side Gap. $\beta_{3k} > 0$ (gender gap in WP quality narrowed post-AI), especially on writing style and contribution framing. ρ_{3k} also narrows (reviewers demand less extra improvement from women post-AI). *Implication:* AI tools are effective equalizers. Policy should promote adoption, especially among female Global South researchers.

Scenario B: AI Equalized Supply but Demand Persists. $\beta_{3k} > 0$ (WP quality gap narrowed) but $\rho_{3k} \approx 0$ (reviewers still demand more from women). *Implication:* The gender tax is primarily demand-driven. AI tools help with manuscripts but cannot fix reviewer bias. Structural reforms (blinding, referee training) are needed.

Scenario C: AI Widened the Geographic Gap. $\beta_{4k} > 0$ (gender gap narrowed for Core authors) but $\beta_{6k} < 0$ (gender gap did not narrow—or widened—for Global South authors). *Implication:* AI tools benefit those who already have access to good infrastructure. The digital divide extends to academic writing. International organizations should subsidize AI writing tools for Global South researchers.

Scenario D: No Effect. All interaction coefficients ≈ 0 . The AI revolution did not change the gender gap. *Implication:* The gap is deeply structural—embedded in norms, networks, and reviewer priors—and does not respond to improvements in writing technology. This is itself a powerful finding: it rules out supply-side explanations.

26 Literature Review Strategy

26.1 The Five Literature Strands

26.1.1 Strand 1: AI as Equalizer in the Workplace

Key papers: Brynjolfsson et al. (2023) (AI compresses skill distribution among customer service agents); Noy and Zhang (2023) (RCT: AI improves writing quality, largest gains for weakest writers); Peng et al. (2023, AI and developer productivity); Dell’Acqua et al. (2023, navigating the jagged frontier). *Your positioning:* “We test the equalization hypothesis in the highest-stakes writing domain: academic publishing, where quality differentials have career consequences.”

26.1.2 Strand 2: Gender Gaps in Peer Review

Key papers: Hengel (2022), Alexander et al. (2023), Card et al. (2020). Frame the AI shock as providing a natural experiment to test their mechanisms: if Hengel’s gap is supply-driven, AI should close it.

26.1.3 Strand 3: Technology Adoption and the Digital Divide

Key papers: Acemoglu and Restrepo (2020) (technology and labor); Goldfarb and Tucker (2019, digital economics); any 2024–2026 papers on AI adoption patterns by geography. *Your positioning:* “If AI adoption is lower among Global South researchers, the AI revolution may widen the gap it was expected to close.”

26.1.4 Strand 4: Natural Experiments from Technology Shocks

Key papers: Hjort and Poulsen (2019, arrival of fast internet in Africa, QJE); Akerman, Gaarder, and Mogstad (2015, broadband internet and labor markets). These are methodological precedents for using technology arrival as a diff-in-diff instrument.

26.1.5 Strand 5: AI Detection and Text Analysis

Key papers: Liang et al. (2023) (GPT detectors biased against non-native writers); Mitchell et al. (2023, DetectGPT); Kirchenbauer et al. (2023, watermarking). *Your*

positioning: you use detection not to police AI use but as a *treatment intensity* measure for economic analysis.

26.2 Critical Differentiation from Existing Work

Table 10: What This Paper Does That Others Cannot

Paper	What they study	Why they cannot answer your question
Brynjolfsson et al. 2023	AI and customer service agents	Not academic writing; no gender gap focus
Noy and Zhang 2023	AI and professional writing (RCT)	Lab setting; not peer review outcomes
Hengel 2022	Gender gap, readability	Pre-AI era; no technology shock
Alexander et al. 2023	Gender gap, review duration	Pre-AI era; no technology shock
Your paper	AI shock $\times$ gender gap	First to combine natural experiment with peer review gender gap

27 Feasibility and Cost

A key advantage of this design: it shares data infrastructure entirely with Agenda 1 and requires no participant recruitment, IRB approval, or multi-year tracking.

Table 11: Incremental Cost of Agenda 3 Beyond Agenda 1

Item	Cost	Notes
Additional AI evaluations (post-AI papers)	\$5,000–8,000	Marginal inference costs
AI-fingerprint score computation	\$1,000–2,000	Perplexity/burstiness analysis
Author survey (subsample, ~200 authors)	\$3,000	AI adoption validation
Total incremental cost	\$9,000–13,000	

28 Timeline

Table 12: Timeline for Agenda 3

Phase	Activity	Timeline
1	Extend Agenda 1 scraping to include post-AI WPs (2023–2025)	Months 2–5
2	AI evaluation of post-AI working papers	Months 5–8
3	AI-fingerprint score construction and validation	Months 6–8
4	Author survey on AI tool adoption (subsample)	Months 7–9
5	Diff-in-diff and event study estimation	Months 9–12
6	Heterogeneity and robustness analysis	Months 11–14
7	Paper writing and submission	Months 13–17

How the Three Agendas Connect

The three agendas form a coherent research program with logical dependencies:

1. **Agenda 2 validates the tool.** Before using AI as a gender-blind benchmark (Agenda 1) or interpreting AI-driven quality changes (Agenda 3), we must know whether the AI itself discriminates. Agenda 2 answers this question.
2. **Agenda 1 documents the problem.** Using the validated AI tool, Agenda 1 quantifies the gender tax across 12 dimensions, identifies where women bear the heaviest burden, and tests whether the tax compounds for Global South researchers. This establishes the *cross-sectional* baseline.
3. **Agenda 3 tests whether the problem is changing.** Using the same data infrastructure as Agenda 1, Agenda 3 exploits the ChatGPT/GPT-4 natural experiment to test whether the gender tax has begun to narrow. This adds *time-series* variation to Agenda 1's cross-sectional analysis, providing the supply/demand decomposition without an RCT.

The Global South dimension runs through all three:

- **Agenda 2:** Tests whether AI discriminates based on non-Western names and Global South affiliations.
- **Agenda 1:** Tests whether the gender tax is amplified at the gender $\times$ geography intersection.
- **Agenda 3:** Tests whether AI tools served as an equalizer or amplifier for Global South researchers.

Shared infrastructure: All three agendas use the same data pipeline (NBER, IZA, WB, SSRN, OpenAlex), the same AI evaluation system (ResearchAI), and the same author classification procedures. Building the infrastructure for one agenda makes the others nearly costless at the margin.

Realistic timeline:

- **Months 1–2:** Pilot studies for Agendas 1 and 2 simultaneously. Validate AI scoring, estimate effect sizes.
- **Months 2–8:** Data collection (shared). Agenda 2 experiments (audit study). Agenda 1 AI evaluation of matched pairs.

- **Months 8–12:** Agenda 2 paper drafted. Agenda 1 analysis. Agenda 3 estimation begins (using Agenda 1's data extended to the post-AI period).
- **Months 12–17:** All three papers drafted. Agenda 2 submitted first (shortest, validates the tool). Agendas 1 and 3 submitted in parallel or sequentially.

Total realistic timeline: 17 months from pilot to submission of all three papers. No IRB, no participant recruitment, no multi-year tracking. All three agendas can be executed by a small team with access to computing resources and the ResearchAI system.

References

- Acemoglu, D. and Restrepo, P. (2020). Robots and jobs: Evidence from US labor markets. *Journal of Political Economy*, 128(6):2188–2244.
- Alexander, D., Gorelkina, O., Hengel, E., and Tol, R. (2023). Gender and the time cost of peer review. *Working paper*.
- Bayer, P. and Charles, K. (2018). Divergent paths: A new perspective on earnings differences between black and white men since 1940. *Quarterly Journal of Economics*, 133(3):1459–1501.
- Bertrand, M. and Mullainathan, S. (2004). Are Emily and Greg more employable than Lakisha and Jamal? A field experiment on labor market discrimination. *American Economic Review*, 94(4):991–1013.
- Biber, D. and Gray, B. (2016). *Grammatical Complexity in Academic English: Linguistic Change in Writing*. Cambridge University Press.
- Blank, R. (1991). The effects of double-blind versus single-blind reviewing: Experimental evidence from the *American Economic Review*. *American Economic Review*, 81(5):1041–1067.
- Bohren, J.A., Imas, A., and Rosenberg, M. (2019). The dynamics of discrimination: Theory and evidence. *American Economic Review*, 109(10):3395–3436.
- Brock, J.M. and De Haas, R. (2022). Discriminatory lending: Evidence from bankers in the lab. *American Economic Journal: Applied Economics*, 15(2):31–68.
- Brynjolfsson, E., Li, D., and Raymond, L. (2023). Generative AI at work. *NBER Working Paper* 31161.
- Card, D., DellaVigna, S., Funk, P., and Iriberri, N. (2020). Are referees and editors in economics gender neutral? *Quarterly Journal of Economics*, 135(1):269–327.
- Checco, A., Bracciale, L., Loreti, P., Pinfield, S., and Bianchi, G. (2021). AI-assisted peer review. *Humanities and Social Sciences Communications*, 8(1):1–11.
- Coffman, K. (2014). Evidence on self-stereotyping and the contribution of ideas. *Quarterly Journal of Economics*, 129(4):1625–1660.
- Das, J., Do, Q.-T., Shaines, K., and Srinivasan, S. (2013). U.S. and them: The geography of academic research. *Journal of Development Economics*, 105:112–130.

- Dressel, J. and Farid, H. (2018). The accuracy, fairness, and limits of predicting recidivism. *Science Advances*, 4(1):eaao5580.
- Ellison, G. (2002). The slowdown of the economics publishing process. *Journal of Political Economy*, 110(5):947–993.
- Exley, C. and Kessler, J. (2022). The gender gap in self-promotion. *Quarterly Journal of Economics*, 137(3):1345–1381.
- Gentzkow, M., Kelly, B., and Taddy, M. (2019). Text as data. *Journal of Economic Literature*, 57(3):535–574.
- Hengel, E. (2022). Publishing while female: Are women held to higher standards? Evidence from peer review. *Economic Journal*, 132(648):2951–2991.
- Hyland, K. (2005). Stance and engagement: A model of interaction in academic discourse. *Discourse Studies*, 7(2):173–192.
- Kleinberg, J., Lakkaraju, H., Leskovec, J., Ludwig, J., and Mullainathan, S. (2018). Human decisions and machine predictions. *Quarterly Journal of Economics*, 133(1):237–293.
- Kline, P., Rose, E., and Walters, C. (2022). Systemic discrimination among large U.S. employers. *Quarterly Journal of Economics*, 137(4):1963–2036.
- Liang, W., Yuksekgonul, M., Mao, Y., Wu, E., and Zou, J. (2023). GPT detectors are biased against non-native English writers. *Patterns*, 4(7):100779.
- Noy, S. and Zhang, W. (2023). Experimental evidence on the productivity effects of generative artificial intelligence. *Science*, 381(6654):187–192.
- Obermeyer, Z., Powers, B., Vogeli, C., and Mullainathan, S. (2019). Dissecting racial bias in an algorithm used to manage the health of populations. *Science*, 366(6464):447–453.
- Raghavan, M., Barocas, S., Kleinberg, J., and Levy, K. (2020). Mitigating bias in algorithmic hiring: Evaluating claims and practices. *ACM Conference on Fairness, Accountability, and Transparency*, 469–481.
- Sarsons, H., Gerarden, K., Cranmer, P., and Heyes, A. (2021). Gender differences in recognition for group work. *Journal of Political Economy*, 129(1):101–147.